Number 3 — Prime factors · HCF and LCM · Ratio

No calculator. Marks in brackets. ★ = harder question.

1 Express 1260 as a product of its prime factors, in index form. [4]

> Divide by primes:
> $1260 = 2 \times 630 = 2 \times 2 \times 315 = 2 \times 2 \times 3 \times 105 = 2 \times 2 \times 3 \times 3 \times 35 = 2 \times 2 \times 3 \times 3 \times 5 \times 7$. So $1260 = 2^2 \times 3^2 \times 5 \times 7$.

2 Find the HCF and the LCM of 48 and 180. [4]

> $48 = 2^4 \times 3$ and $180 = 2^2 \times 3^2 \times 5$. HCF — shared primes, lowest powers: $2^2 \times 3 = 12$. LCM — all primes, highest powers: $2^4 \times 3^2 \times 5 = 720$. (Check: $12 \times 720 = 8640 = 48 \times 180$ ✓)

3 Two teachers set regular tests. Ms Avery's tests always have 20 questions and Mr Baig's always have 36. In one term both classes answer the same total number of questions. [4]

a What is the smallest total number of questions that makes this possible?

b How many tests does Ms Avery's class take in that case?

> **a** The total is a multiple of both 20 and 36 — the smallest is the LCM: $20 = 2^2 \times 5$, $36 = 2^2 \times 3^2$, LCM $= 2^2 \times 3^2 \times 5 = 180$ questions.
> **b** $180 \div 20 = 9$ tests (Mr Baig's class takes 5).

4 Nadia has 180 pens and 300 notebooks to pack into identical goody bags, using everything. What is the greatest number of bags she can make, and what goes in each bag? [4]

> The number of bags divides both 180 and 300 — the greatest is the HCF: $180 = 2^2 \times 3^2 \times 5$, $300 = 2^2 \times 3 \times 5^2$, HCF $= 2^2 \times 3 \times 5 = 60$ bags. Each bag: $180 \div 60 = 3$ pens and $300 \div 60 = 5$ notebooks.

5 Three cousins aged 45, 18 and 9 share €720 in the ratio of their ages. How much does each receive? [4]

> Simplify the ratio first: $45 : 18 : 9 = 5 : 2 : 1$ (÷9). $5 + 2 + 1 = 8$ parts; $720 \div 8 = 90$ per part. Shares: Math input error, Math input error, Math input error (total 720 ✓).

6 Theo finds that 45 ml of cordial mixed with 270 ml of water makes a drink he likes. [5]

a Write the ratio cordial : water in its simplest form.

b Theo uses 6 litres of water to make the same strength of drink. What total volume of drink does he make?

> **a** $45 : 270 = 1 : 6$ (÷45).
> **b** Water is 6 parts, so one part is $6 \div 6 = 1$ litre of cordial. Total $= 1 + 6 = 7$ litres.

7 Which ratio is larger, $5 : 3$ or $17 : 10$? Show how you decide. [2]

> Write each as a unit ratio $n : 1$ — divide by the second number: $5 : 3 = 1.67 : 1$ and $17 : 10 = 1.7 : 1$. So $17 : 10$ is the larger ratio (by a little).

8 Two gear wheels have 24 and 40 teeth and are meshed together. A mark is painted where they touch. What is the smallest number of turns of the 24-tooth wheel before the two marks meet again? [3]

> The marks meet again after a number of teeth that is a multiple of both 24 and 40: LCM $= 2^3 \times 3 \times 5 = 120$ teeth. That is $120 \div 24 = 5$ turns of the small wheel (and 3 of the large).

Total: 30 marks

Check: "greatest number of equal groups" → HCF · "next time they coincide" → LCM · simplify a ratio before sharing · the HCF × LCM of two numbers equals their product.